

# Economics 154: The Global Economy: Trade, Finance, and Policy

## Lecture 2 Notes: Scarcity, Incentives, Marginal Reasoning, Evidence, and Models

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### Purpose and Learning Objectives

The first class introduced the course and showed why economics helps us interpret major events. This lecture now builds the core toolkit students will use throughout the semester. The central idea is that scarcity forces choices, choices have opportunity costs, incentives change behavior, and good economic reasoning asks what happens at the margin. We then extend this framework to evidence: economists distinguish positive from normative claims, correlation from causation, and complicated reality from the simplified models used to understand it.

By the end of the lecture you should be able to:

- explain scarcity and opportunity cost;
- identify how incentives change behavior;
- apply marginal benefit–marginal cost reasoning;
- distinguish positive and normative statements;
- explain why correlation need not imply causation;
- explain why economists use models and the *ceteris paribus* assumption;
- apply these tools to real economic questions and news stories.

## 1 Scarcity: The Starting Point

Economics begins with scarcity. Scarcity means that available resources are limited relative to the many things people, firms, and societies want. Scarcity is not the same thing as poverty. Even wealthy individuals, governments, firms, and countries face scarcity because time, attention, labor, land, money, energy, water, and institutional capacity are limited.

Scarcity is therefore the reason choices exist. A student has only so many hours in a day. A household has a budget. A firm has limited workers and equipment. A government has limited revenue and political attention. A country has limited productive capacity.

### Scarcity is not only about money

- **Time:** students cannot study, work, sleep, exercise, and socialize at the same hour.
- **Income:** households cannot spend the same dollar on rent, food, travel, and savings.
- **Labor:** firms cannot hire the same worker for two different jobs at the same time.
- **Natural resources:** water, land, minerals, and energy are limited.
- **Attention:** governments and organizations cannot focus intensely on every problem at once.

## 2 Real-World Example: Water Scarcity and the Panama Canal

A useful global example comes from the Panama Canal. The canal depends on freshwater to operate its locks. When rainfall is unusually low, water becomes a binding constraint for both canal operations and the surrounding population. The resulting restrictions illustrate scarcity in physical rather than monetary terms.



Figure 1: Illustrative container-ship image used in the original Lecture 2 notes. The Panama Canal example shows how physical scarcity can affect global trade.

### Think Like an Economist

What is the scarce resource? What alternative uses compete for it? Who bears the opportunity cost when transit capacity is restricted?

## 3 Opportunity Cost

Once scarcity forces a choice, the next question is what must be given up.

### Definition

The **opportunity cost** of a choice is the value of the next-best alternative forgone.

Opportunity cost is broader than an explicit monetary payment. A student attending college pays tuition, but also gives up time that could have been used to work. A government that spends an additional billion dollars on one program gives up the best alternative use of those resources.

### Explicit and implicit costs

- **Explicit costs** involve direct payments such as tuition, wages, fuel, or rent.
- **Implicit costs** are valuable alternatives forgone, such as time, foregone earnings, or use of owned resources.

## 4 Example: Education, Earnings, and Opportunity Cost

Education provides a useful example because its benefits and costs occur over time. More schooling is associated with higher average earnings, but attending school also involves tuition, time, effort, and foregone earnings.

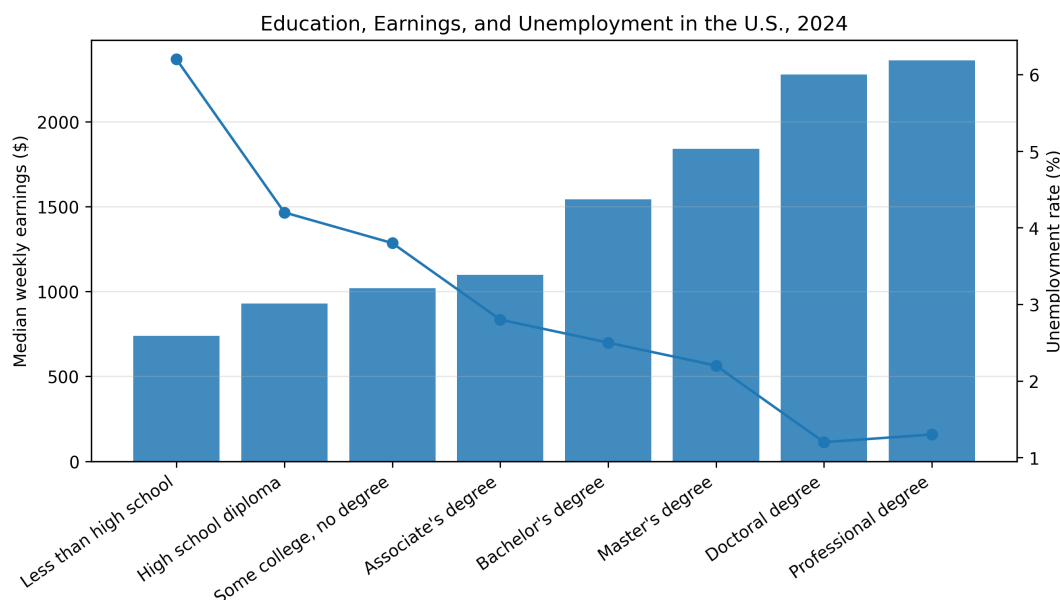


Figure 2: The original Lecture 2 BLS “Education Pays” figure. The association between education and earnings motivates opportunity-cost and evidence questions.

The graph is informative, but economic reasoning asks two separate questions. First, is additional education worth its opportunity cost for a particular individual? Second, how much of the observed earnings difference is actually caused by education? We return to the second question when discussing correlation and causation.

## 5 Incentives

An incentive changes the benefits or costs associated with an action. When incentives change, behavior often changes as well.

Examples include:

- higher wages may attract more applicants;
- a tax on cigarettes raises the private cost of smoking;
- a subsidy for solar panels lowers the private cost of installation;

- congestion tolls make driving into a crowded area more expensive.

The important point is not that people respond mechanically to every incentive. Rather, economists expect behavior to respond when the relative benefits and costs of alternatives change.

## 6 Real-World Example: Congestion Pricing

Congestion pricing illustrates incentives clearly. When entering a crowded area becomes more expensive, some travelers may change the time of their trip, use public transportation, share a vehicle, or avoid the trip.



Figure 3: Illustrative congestion-pricing image used in the original Lecture 2 notes.

The economic mechanism can be summarized as:

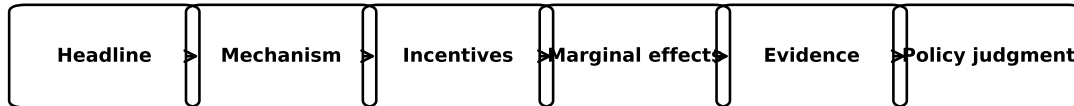
Higher private cost of driving  $\rightarrow$  behavioral response  $\rightarrow$  changes in traffic volume.

This does not by itself tell us whether the policy is desirable. That requires considering benefits, costs, distributional effects, and evidence.

## 7 Thinking Like an Economist

Economic discussions often begin with a headline and quickly become arguments about whether something is good or bad. Economists try to slow that process down.

## Thinking Like an Economist



**Describe and explain first; evaluate policy only after the mechanism and evidence are clear.**

Figure 4: A disciplined sequence for moving from a headline to economic interpretation.

For a policy or headline, ask:

1. What decision is actually being made?
2. Compared with what alternative?
3. What mechanism connects the action to the outcome?
4. Which incentives change?
5. Who changes behavior?
6. What happens at the margin?
7. What evidence would tell us whether the mechanism is important?
8. Only then: what should policy do?

## 8 Marginal Thinking

Economists rarely treat decisions as all-or-nothing. They ask whether doing a little more or a little less improves the outcome.

The word *marginal* means **additional**. Examples include:

- one additional hour of studying;
- one additional worker hired;
- one additional flight;

- one additional dollar of spending;
- one additional quarter-point change in an interest rate.

### Average versus marginal thinking

“College graduates earn more on average” is an average statement. “Would one additional year of education raise this student’s future earnings enough to justify its cost?” is a marginal question.

Likewise, “exercise is good for health” does not tell us whether a fifth additional hour of exercise today is worthwhile. Marginal reasoning recognizes that benefits and costs depend on where we currently are.

### Student example

Suppose you have already studied for three hours. The relevant choice is not studying versus never studying. It is three hours versus four hours. The additional benefit of the fourth hour must be compared with its additional opportunity cost.

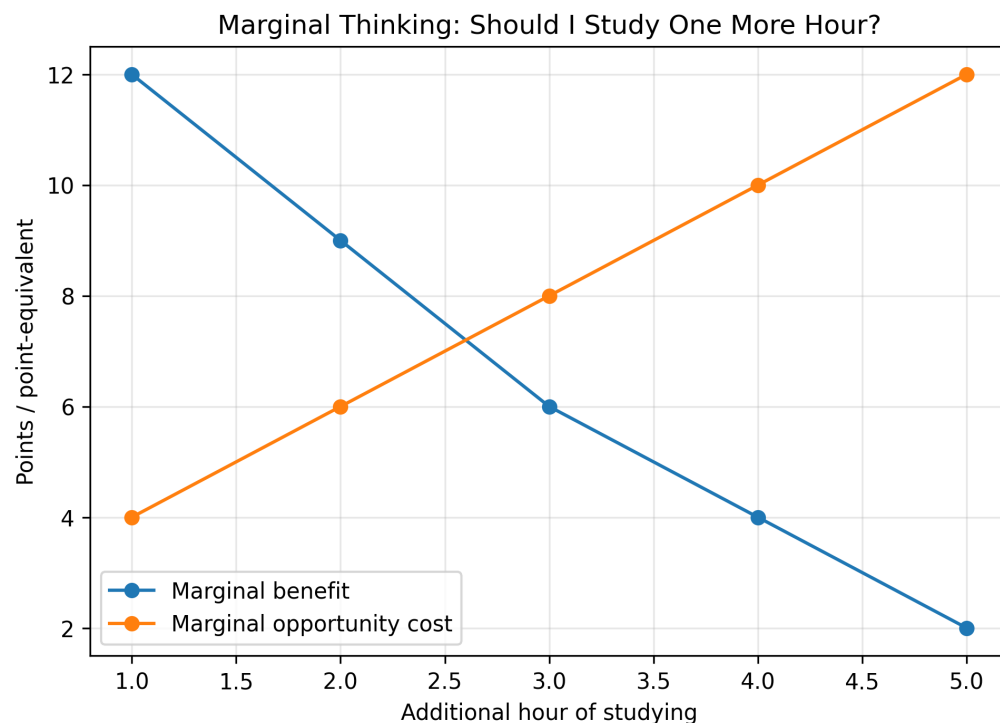


Figure 5: The original Lecture 2 study-time figure illustrates diminishing gains from additional study time.

## 9 Marginal Benefits, Marginal Costs, and the Decision Rule

The **marginal benefit** is the additional benefit from one more unit of an activity. The **marginal cost** is the additional cost of that unit.

Marginal benefits often decline as an activity expands, while marginal costs may rise as increasingly valuable alternatives are sacrificed.

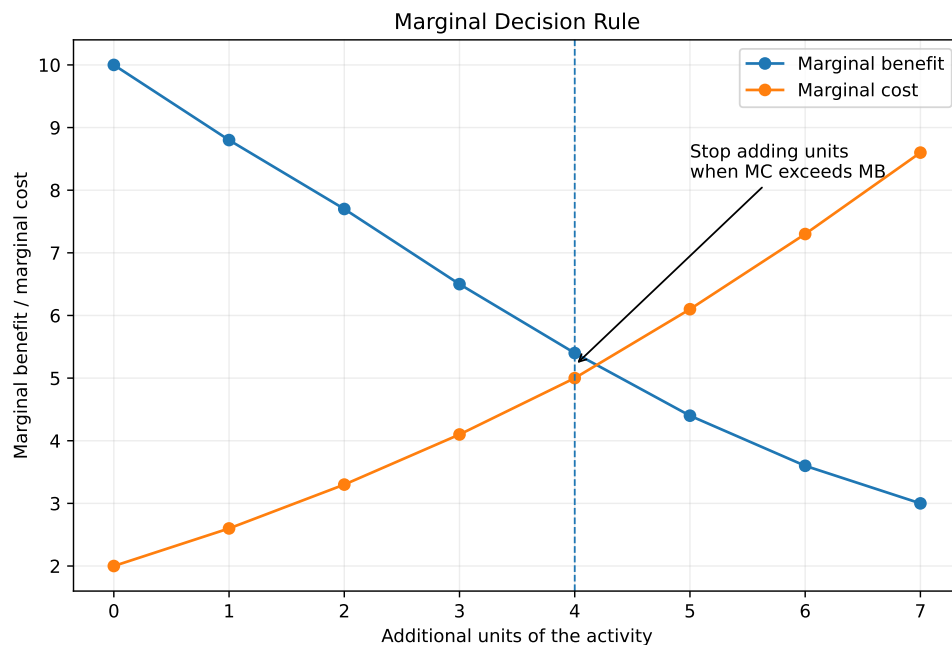


Figure 6: A simple marginal-benefit and marginal-cost representation. Continue expanding an activity while the additional benefit is at least as large as the additional cost.

The basic decision rule is:

Continue while  $MB \geq MC$ .

Once marginal cost exceeds marginal benefit, an additional unit reduces net benefit.

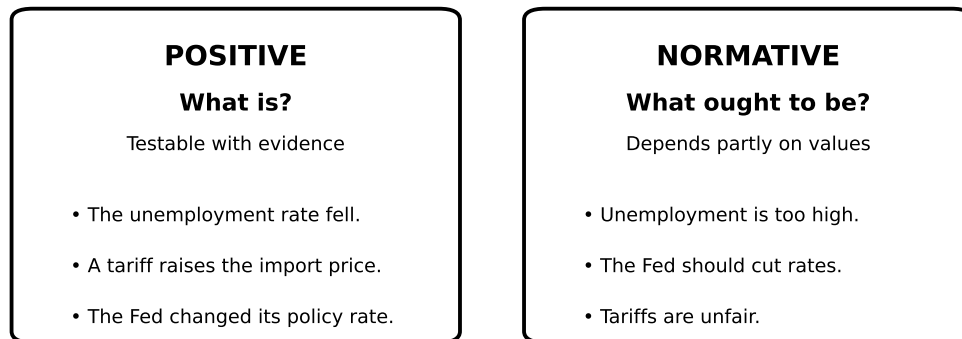
### Example: a central bank decision

Suppose inflation remains elevated. The relevant question is not whether “high rates” or “low rates” are good. It is whether an additional policy-rate increase produces benefits larger than its costs. Possible benefits include lower inflation and more stable expectations; possible costs include slower demand, weaker investment, and employment losses. The logic is marginal.

## 10 Positive and Normative Economics

A crucial step in economic reasoning is separating descriptions of the world from judgments about what ought to happen.

### Positive and Normative Economics



**Evidence informs both—but evidence alone cannot settle a value judgment.**

Figure 7: Positive statements are, in principle, testable with evidence. Normative statements depend partly on values and social objectives.

#### Definition

**Positive economics** asks “What is?” and studies claims that can, at least in principle, be confronted with evidence.

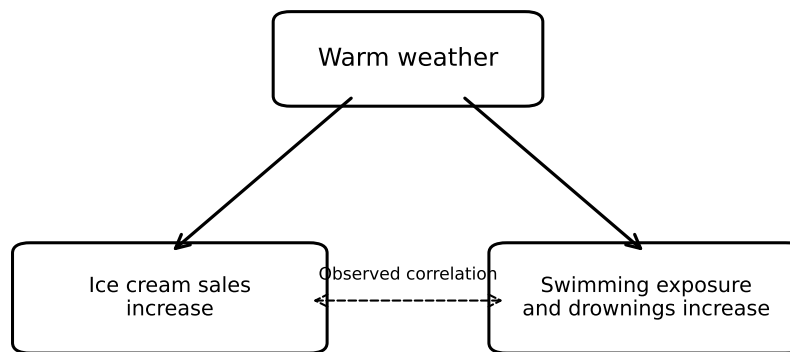
**Normative economics** asks “What ought to be?” and involves values, ethical judgments, or preferences.

For example, “a tariff raises the domestic price of imported goods” is a positive claim. “Tariffs are unfair” is normative. Evidence can inform the normative debate by clarifying consequences, but evidence alone cannot determine how society should weigh efficiency, equity, liberty, security, or environmental protection.

## 11 Correlation and Causation

Two variables are **correlated** if they tend to move together. One variable **causes** another only if changing the first directly changes the second.

### A Third Variable Can Generate a Correlation



**Correlation between two variables does not establish that one causes the other.**

Figure 8: Warm weather can raise both ice-cream sales and swimming exposure. Their correlation does not imply that ice cream causes drownings.

A third variable may generate an observed correlation. Reverse causality is another possibility. Selection can also matter.

## College and earnings revisited

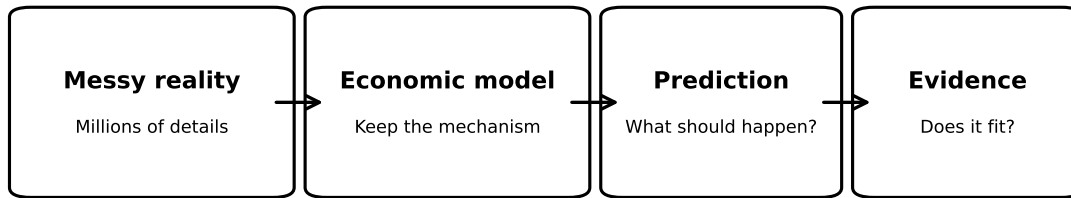
The BLS education graph shows that people with more education have higher average earnings. That is an important fact. But it does not, by itself, prove that the entire difference is caused by schooling. Ability, family background, motivation, field of study, work experience, and local labor-market conditions may also matter.

This is why empirical economics devotes so much effort to research designs that isolate causal effects. Later in the course we will see simple examples of how economists use natural experiments and other sources of plausibly exogenous variation.

## 12 Why Economists Build Models

The real world contains too many details to analyze everything at once. Economists therefore use models: simplified representations designed to isolate mechanisms.

## Why Economists Build Models



**Simplification is useful when it isolates the causal mechanism we want to understand.**

Figure 9: Economic models deliberately simplify reality so that a mechanism can generate predictions that can be compared with evidence.

A subway map is useful precisely because it omits buildings, trees, and most streets. The simplification is not a defect if the purpose is to navigate the subway. Economic models work similarly.

## 13 The Production Possibilities Frontier

The Production Possibilities Frontier (PPF) is an early example of an economic model. It describes the combinations of two outputs that can be produced using available resources and technology.

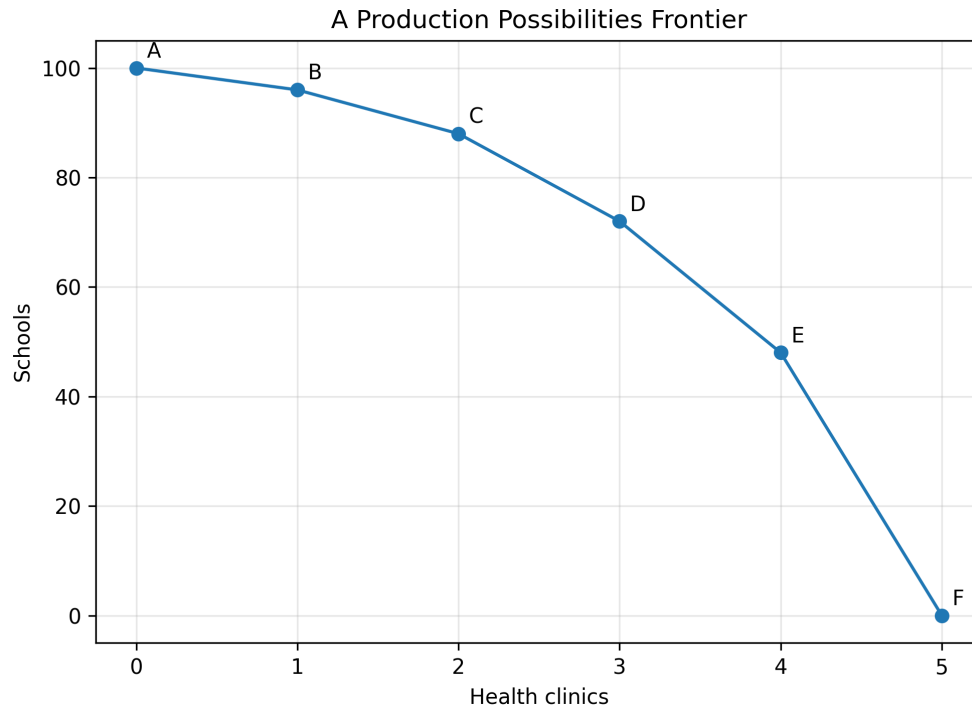


Figure 10: The original Lecture 2 PPF figure. Moving along the frontier illustrates scarcity and opportunity cost.

A point on the frontier uses resources fully. A point inside the frontier indicates unused or inefficiently used resources. A point beyond the frontier is not currently feasible. Moving along the frontier illustrates opportunity cost: producing more of one output requires giving up some of the other.

The bowed-out shape of a typical PPF reflects increasing opportunity cost when resources are progressively shifted toward uses for which they are less well suited.

## 14 *Ceteris Paribus*: Holding Other Things Constant

Models frequently use the assumption *ceteris paribus*: “all other relevant things held constant.”

## Ceteris Paribus: Isolate One Change

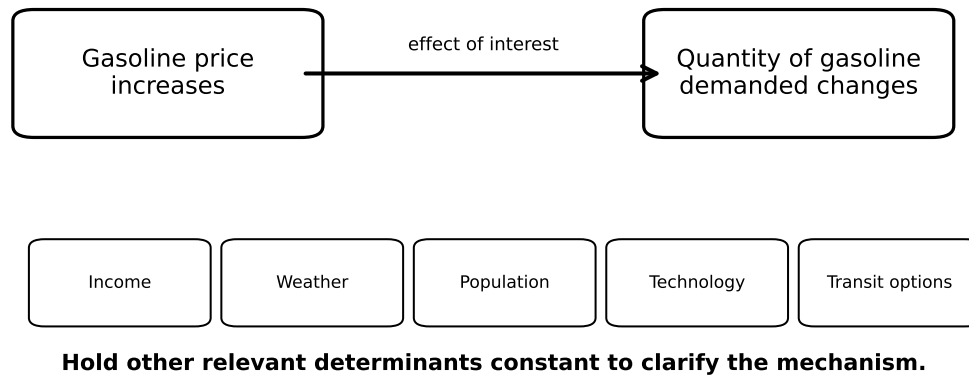


Figure 11: Ceteris-paribus reasoning isolates the effect of one change while holding other relevant determinants fixed.

Suppose gasoline prices rise while income, weather, population, technology, and transit options are also changing. It becomes difficult to isolate what higher gasoline prices did. Economists first simplify the question: what would happen if the price changed while the other relevant determinants remained fixed?

The assumption does not claim that the real world literally freezes. It is an analytical device for understanding one mechanism at a time.

## 15 What Makes Economic Thinking Useful?

Economic thinking disciplines intuition. Instead of moving directly from a headline to an opinion, ask:

- What is scarce?
- What choice is being made?
- What is the opportunity cost?
- What incentives change?
- What are the marginal benefits and marginal costs?
- Is the claim positive or normative?

- Is the evidence causal or merely correlational?
- What model isolates the mechanism?
- What must be held constant?
- Who gains, who loses, and how would we know?

## Takeaways

1. Scarcity makes tradeoffs unavoidable.
2. Opportunity cost is the value of the next-best alternative forgone.
3. Incentives alter behavior by changing relative benefits and costs.
4. Most economic choices are best analyzed at the margin.
5. Positive claims and normative judgments are different.
6. Correlation is not automatically causation.
7. Models simplify reality in order to isolate mechanisms.
8. *Ceteris paribus* helps economists identify the effect of one change at a time.

## Preview: Markets

We now have the basic toolkit required to understand markets. The next lecture applies these ideas to one of economics' most influential models:

### How are prices determined?

The answer begins with demand, willingness to pay, and consumer behavior.

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